

Study on interfacial trap location induced subthreshold slope degradation extracted by random telegraph noise for high-k/metal gate FinFET devices

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ABSTRACT

In this work, constant voltage stress (CVS) induced interfacial trap and its effect on subthreshold slope (SS) degradation were studied for the first time by the random telegraph noise (RTN) technique. Part of the generated traps were located in the interfacial layer (IL) under CVS. Both the trap location and trapping number show a large impact on SS degradation, and it could be demonstrated by RTN. If the trap depths are similar, more traps cause more severe SS degradation. If the trap depths are different, shallow trap shows larger impact on SS degradation than that of deep trap even if the trapping charges of deep trap are more than that of shallow trap.

1. Introduction

FinFET is the most promising candidate for device scaling due to their superior control of the channel. To deal with the scaling, high-k/metal gate (HK/MG) has been in production beyond 28 nm [1]. Device reliability is a fundamental issue for scaling process, and constant voltage stress (CVS) is one of the major reliability concerns for HK/MG device. CVS induced threshold voltage (V_{TH}) degradation, which suggests that traps are trapped in bulk high-k gate dielectric, is still an important reliability problem [2,3]. However, interface damage is rarely discussed in prior studies. Random telegraph noise (RTN) is a special case of Generation-Recombination noise, which displays as discrete switching events in the time domain [4]. The impact of RTN is getting more important since the variations of atomic level can affect the drain current in sub-20 nm devices [5]. It provides unique opportunity for directly measuring the response of an FET to a single charge [6], which is attributed to trapping/detrapping events caused by an individual interface defect near the Si/SiO₂ interface, and the measured RTN signal only displays two discrete current levels as a function of time associated with capture/emission at a single defect site [7–9]. In order to accurately analyze RTN signal in high-k gate dielectric devices, it is necessary to figure out whether an oxide trap is located in the high-k layer or in the interfacial layer (IL) [10]. The bias-voltage dependence of the capture and emission times allows one to determine the location of the traps [11–13], and becomes important in determining the performance and reliability [14].

Sometimes, degradation of subthreshold slope (SS) could be observed after CVS. It is suggested that some of the traps were generated close to the Si/IL interface. Drain current RTN (I_D -RTN) was utilized in this work to extract the traps and find the connection between SS degradation and trap location.

2. Experimental

The N-type tri-gate FinFET was fabricated on a bulk-Si wafer. A thin (~1 nm) interfacial SiO₂ layer and 2-nm-thick Hafnium based high-k gate material was grown as a gate dielectric and followed by stack scavenge inserted TaN/TiN/Ti metal film gate electrode. The channel length, fin width and fin height of the devices are 20 nm, 10 nm and 42 nm, respectively.

Table 1 shows the stress condition utilized in this work. After the measurement of basic electrical properties and RTN on fresh devices at room temperature, the devices were stressed under constant voltage stress condition of 2.3 V, 2.6 V, 2.9 V, 3.0 V, 3.1 V and 3.2 V for 20 s step by step, and each stress step is followed by the measurement of basic electrical properties and RTN. A semiconductor parameter analyzer (Agilent-B1500A) was utilized to complete the measurement in this work.

3. Results and discussions

The inset of Fig. 1 shows the I_D - V_G curves for FinFET devices under

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Table 1
CVS conditions.

Stress step	Stress condition
St.0	CVS 2.3 V, 20 s
St.0'	St. 0 + CVS 2.6 V, 20 s
St.1	St. 0' + CVS 2.9 V, 20 s
St.2	St. 1 + CVS 3.0 V, 20 s
St.3	St. 2 + CVS 3.1 V, 20 s
St.4	St. 3 + CVS 3.2 V, 20 s

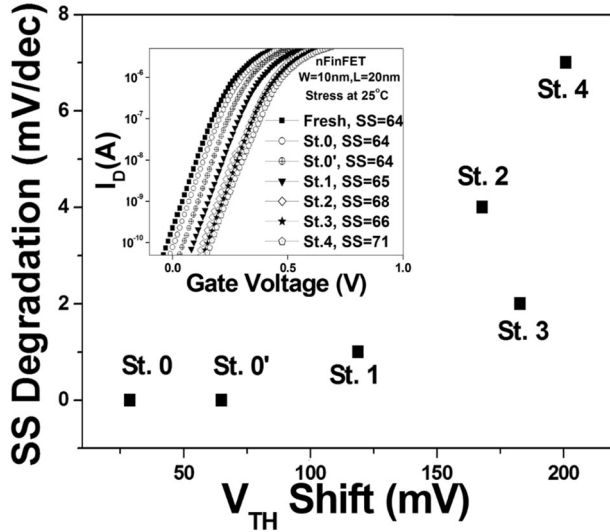


Fig. 1. The extracted V_{TH} shift versus SS degradation. SS degradation is found not only dependent of V_{TH} shift. The inset shows the I_D - V_G curves for FinFET devices under various constant voltage stresses.

various constant voltage stresses, and the extracted V_{TH} shift versus SS degradation is also shown in Fig. 1. After stress step of St.0 and St.0', V_{TH} shift from 29 mV to 65 mV could be observed without SS degradation. It is supposed that if the traps were generated in bulk high-k dielectric or oxide away from the interface, V_{TH} shift with negligible SS variation could be shown. On the other hand, if the traps were located close to Si/SiO₂ interface, both V_{TH} shift and SS degradation could be found. As a result, it is believed that there should be rare traps generated in interfacial layer but more traps generated in bulk dielectric under low stress voltage such as stress step of St.0 and St.0'. After stress step of St.1, both V_{TH} shift and SS degradation were obvious, indicating not only more traps in bulk dielectric but also traps generated at Si/SiO₂ interface. It is interesting to find that stress step of St.3 shows larger V_{TH} shift but smaller SS degradation than that of St.2. It is suggested that there might be more traps generated after stress step of St.3 while the trap location might be much closer to Si/SiO₂ interface after stress step of St.2. It is supposed that SS degradation would be affected by both the location and the number of traps generated in dielectric.

In order to figure out the generation of traps that corresponds to SS degradation, RTN measurements were utilized in this work. Fig. 2 shows the RTN signal measured after each stress step. There was no RTN signal observed for fresh devices (Fig. 2a) and for devices after stress step of St.0 (Fig. 2b) and St.0' (Fig. 2c). It is suggested that there is no interfacial defect that could be detected by RTN under this stress condition even though a large V_{TH} shift could be observed. A two-level RTN signal could be detected after stress step of St.1 when $V_{DS} = 0.2$ V (Fig. 2d). With stress step in progress, two-level RTN signals could also be detected after St.2 when $V_{DS} = 0.3$ V (Fig. 2e). After St.3, two two-level RTN signals could also be detected when $V_{DS} = 0.3$ V (Fig. 2f) and $V_{DS} = 0.1$ V (Fig. 2g), expressed as St.3(a) and St.3(b), respectively. And a four-level RTN signal could be observed after St.4 (Fig. 2h). Prior

study has indicated that a single trap exhibits a two-level I_D waveform, while a four-level RTN signal can be caused by two overlapping traps. In fact, if there are two traps that are not overlapping along the channel, two two-level RTN signals could be found under different drain bias [15].

In order to find out the depth of the traps, $\ln(\tau_c/\tau_e)$ versus gate voltage was plotted and shown in Fig. 3. The value of $d\ln(\tau_c/\tau_e)/dV_G$ determines the depth of trap in gate dielectric [10]. If the trap is located close to the interface, it is easier for carrier to be captured, resulting smaller τ_c/τ_e value at fixed gate voltage. The variation of capture/emission time extracted from RTN signal is relatively slight with various gate voltages. As a result, the slope of $d\ln(\tau_c/\tau_e)/dV_G$ is relatively slow. On the other hand, if the generated trap is far away from interface, it is more difficult for carrier to be captured, resulting larger τ_c/τ_e value at fixed gate voltage, corresponding to a steeper slope of $d\ln(\tau_c/\tau_e)/dV_G$. For the device used in this work, an oxide trap is found to be located in IL if the time constant variation rate of $d\ln(\tau_c/\tau_e)/dV_G$ is more negative than -28.3 V⁻¹, and is located in high-k layer if $d\ln(\tau_c/\tau_e)/dV_G$ is less negative than -28.3 V⁻¹. The extracted time constant variation rate indicates that the detected traps of two-level RTN are located in IL. The position of RTN traps can be determined from formula (1) and (2) [10].

$$X_T = \left(T_{OX1} + \frac{\epsilon_{OX1}}{\epsilon_{OX2}} T_{OX2} \right) \left(\frac{K_B T}{q} \frac{d \ln(\tau_c/\tau_e)}{dV_G} + \frac{d\psi_s}{dV_G} \right) / \left(\frac{d\psi_s}{dV_G} - 1 \right) \quad (1)$$

$$Y_T = \left[\frac{K_B T}{q} \frac{1}{x_{T1}} \left(T_{OX1} + \frac{\epsilon_{OX1}}{\epsilon_{OX2}} T_{OX2} \right) \ln \frac{\ln(\tau_c/\tau_e)_f}{\ln(\tau_c/\tau_e)_r} + V_{DS,r} \right] / \left[\frac{V_{DS,f} + V_{DS,r}}{L_{eff}} \right] \quad (2)$$

It can be shown that after stress step of St.1, a trap (trap 1) was observed in IL while its X_T and Y_T/L_{eff} are extracted to be 0.874 nm referenced from IL/Si interface and 0.68 from the edge of the source, respectively. With stress step in progress, a trap could be detected after the stress step of St.2 (trap 2) with its X_T and Y_T/L_{eff} being 0.57 nm and 0.56, respectively. A shallow trapping depth can be found due to the random generation of defect. When the stress step is under St.3, two two-level RTN signals could be found due to two non-overlapped traps. One of the traps is located at $X_T = 0.798$ nm and $Y_T/L_{eff} = 0.51$ (trap 3a) while the other trap is located at $X_T = 0.836$ nm and $Y_T/L_{eff} = 0.77$ (trap 3b). The four-level RTN signal could be observed after stress step of St.4 only at bias of $V_G = 0.996$ V and $V_{DS} = 0.1$ V, which indicates that two overlapped traps exist. There is no RTN signal that could be detected at any other measurement bias, showing that there are more traps generated after St.4 than other stress steps.

It is well known that interface state generation results in SS degradation [16]. The degradation is suggested due to capture and emission of carriers from the interface traps. On the other hand, if the generated trap is further away from interface, a minor influence of SS degradation is reported [3]. As a result, it is suggested that the depth of generated trap would strongly affect SS degradation.

In order to verify our suggestion, the extracted trap locations are illustrated as shown in Fig. 4. After stress step of St.1, little SS degradation (shown in Fig. 1) is caused by the single trap (trap 1) with a depth of 0.874 nm. As stress goes on, depth of trap 2 under stress step of St.2 is located on 0.57 nm. Compared to trap 1, trap 2 is much closer to Si/IL interface. SS degradation with St.2 shows more serious than that with St.1, which is because that SS degradation corresponds with interface damage. As a trap is closer to Si/IL interface, it is easier for carrier trapping/detrapping. As a result, more severe SS degradation could be observed due to the shallow trap. As stress step is under St.3, two non-overlapped traps could be extracted by RTN. The trap depths for trap 3a and trap 3b are 0.798 nm and 0.836 nm, respectively. The trap locations of these two traps are similar to that of trap 1. However, SS degradation of St.3 is a little more severe than that of St.1, due to more traps located in IL. On the other hand, SS degradation of St. 3,

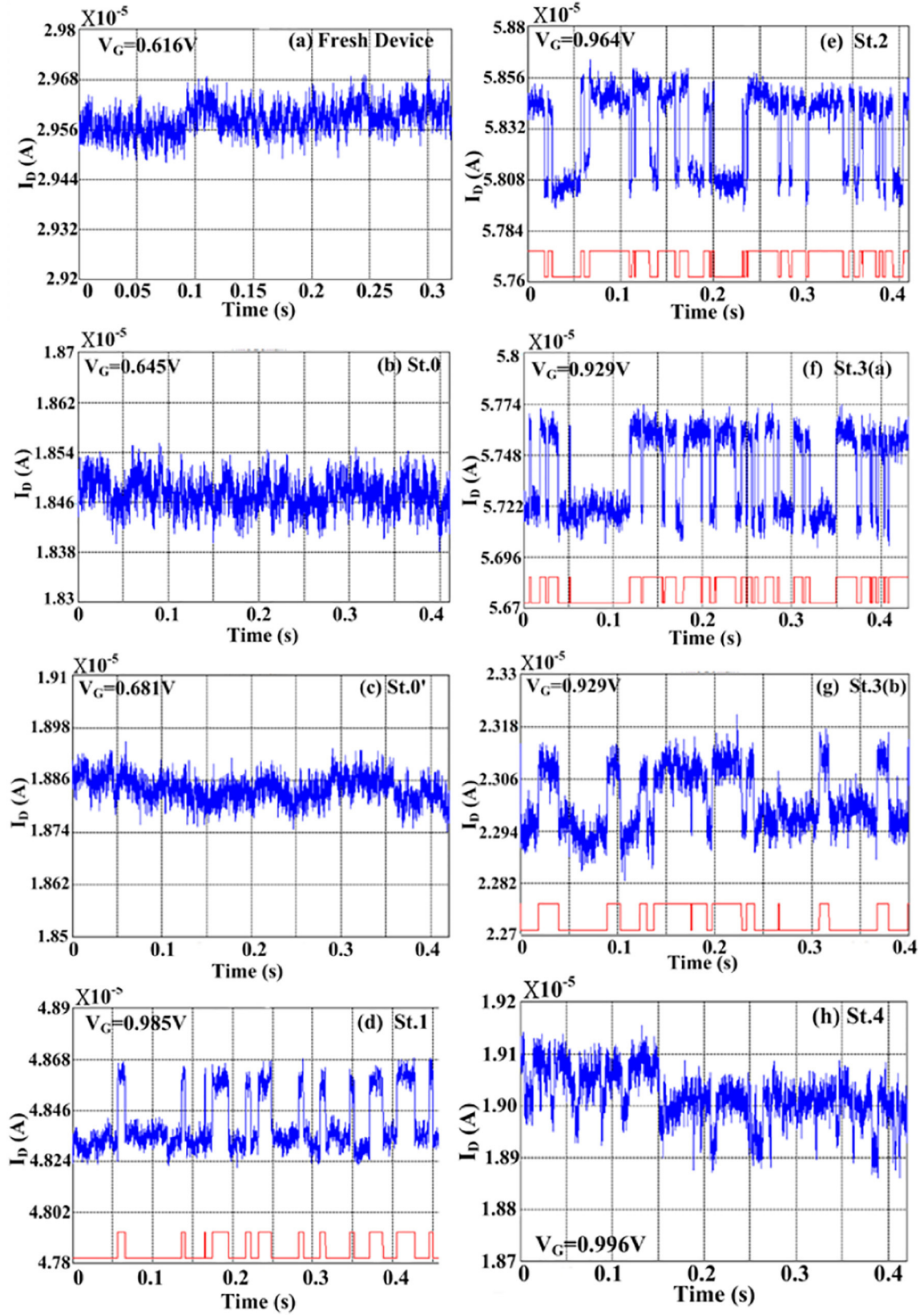


Fig. 2. Random Telegraph Noise signal measured after each stress step.

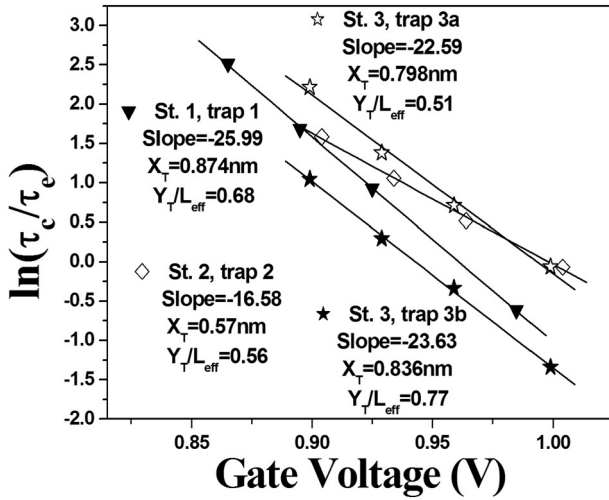


Fig. 3. Plot of $\ln(\tau_c/\tau_e)$ versus gate voltage. The effective depth location of traps could be extracted.

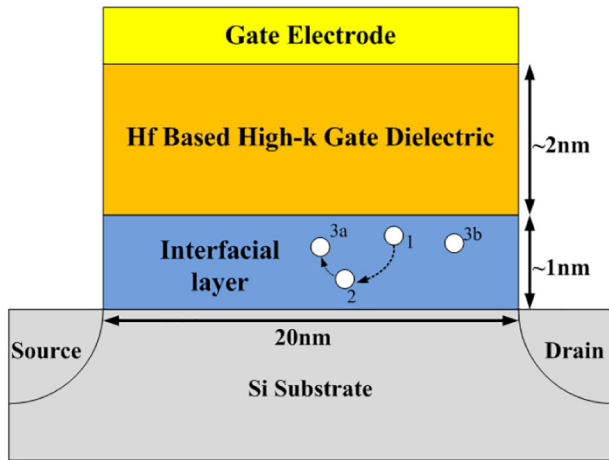


Fig. 4. The illustration of trap locations in each stress step.

which has two traps, is slighter than that of St. 2 which has only one trap. It is suggested that when comparing the impact of SS degradation, one shallow trap shows larger impact on SS degradation than that of two deep traps. As the device was under stress step of St.4, the four-level RTN signal indicates that the generated traps in gate dielectric is much more than that in prior stress steps. As a result, more severe SS degradation and larger V_{TH} shift could be expected, which is corresponding to the data shown in Fig. 1.

4. Conclusion

Constant voltage stress induced random traps generated in interfacial layer were extracted by RTN in this work. It is suggested that if there is no trap generated in IL, constant voltage stress only introduces V_{TH} shift and no SS degradation could be observed. SS degradation was found when a trap was located in interfacial layer, and the trap location could be extracted by RTN method. Device with more traps shows more serious SS degradation if the trap depths are similar. On the other hand, trap depth shows larger impact on SS degradation. It shows more severe SS degradation on devices with one shallow trap than those of two deep traps. The connection between trap location and SS degradation could be clearly observed by using RTN.

CRediT authorship contribution statement

Yi-Lin Yang: Conceptualization, Methodology, Investigation, Writing - original draft, Writing - review & editing. **Wenqi Zhang:** Writing - original draft, Writing - review & editing. **Yu-Lin Chen:** Investigation. **Wen-Kuan Yeh:** Conceptualization, Resources, Supervision.

Declaration of competing interest

We confirm that the manuscript has been read and approved by all named authors and that there are no other persons who satisfied the criteria for authorship but are not listed. We further confirm that the order of authors listed in the manuscript has been approved by all of us.

We confirm that we have given due consideration to the protection of intellectual property associated with this work and that there are no impediments to publication, including the timing of publication, with respect to intellectual property. In so doing we confirm that we have followed the regulations of our institutions concerning intellectual property. We further confirm that any aspect of the work covered in this manuscript that has involved either experimental animals or human patients has been conducted with the ethical approval of all relevant bodies and that such approvals are acknowledged within the manuscript.

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